

Qn	Answer	Solution
16	C	Power received by solar panel = Intensity at distance from Earth $\times$ area of panel $\Rightarrow P = I \times A$ $1.3 \times 10^3 = \frac{P_s}{4\pi (1.5 \times 10^{11})^2} \times 1.0 \text{ -----(1)}$ $1.3 \times 10^3 = \frac{P_s}{4\pi (2.3 \times 10^{11})^2} \times A \text{ -----(2)}$ (1) = (2), solving $\Rightarrow A = 2.4 \text{ m}^2$ P_s: power from source, Sun
17	D	Using $A = A_0 \cos \theta$